

Global dynamics of population, land use, and inequality

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Overview

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Datasets

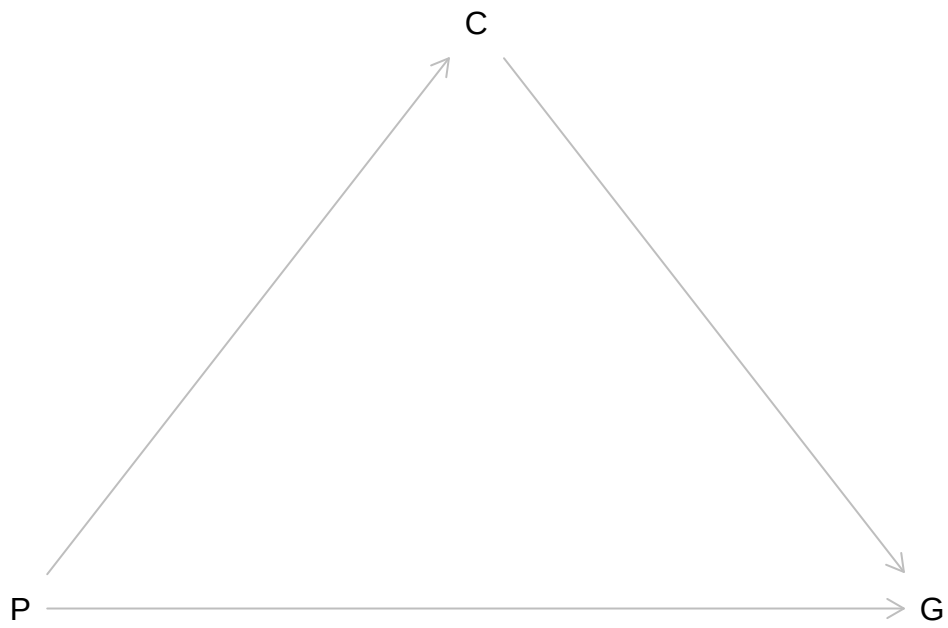
...

Causal model

Static DAG:

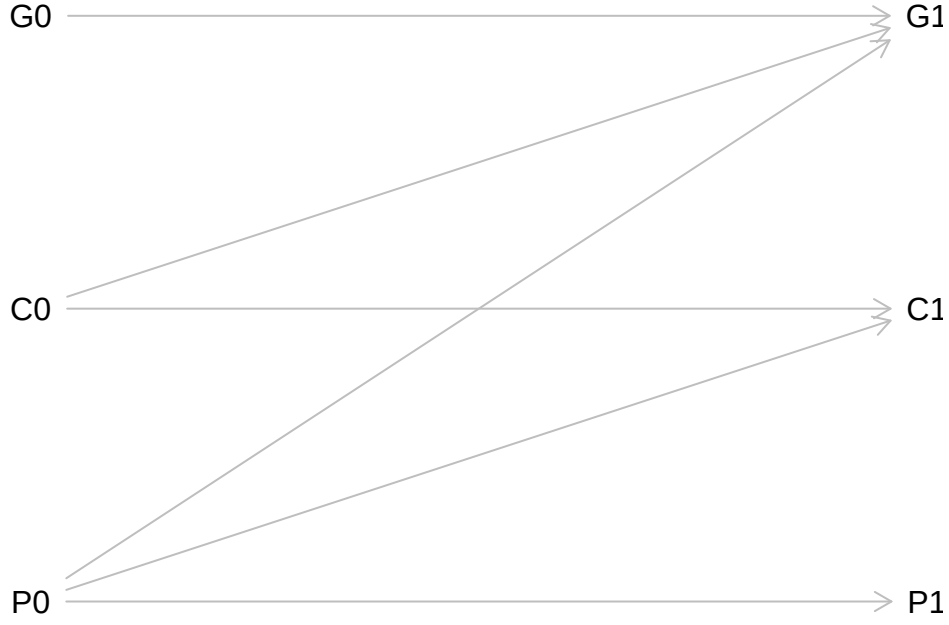
```
library(dagitty)

dagitty('dag{
  P [pos = "0,0"]
  C [pos = "1,-1"]
  G [pos = "2,0"]
  G <- P ;
  G <- C ;
  C <- P ;
}') |> plot()
```



Dynamic DAG:

```
dagitty('dag{  
  P0 [pos = "0,0"]  
  P1 [pos = "1,0"]  
  C0 [pos = "0,-1"]  
  C1 [pos = "1,-1"]  
  G0 [pos = "0,-2"]  
  G1 [pos = "1,-2"]  
  P1 <- P0 ;  
  C1 <- C0 ;  
  G1 <- G0 ;  
  C1 <- P0 ;  
  G1 <- P0 ;  
  G1 <- C0 ;  
}') |> plot()
```



Statistical model

Likelihoods:

$$P \sim \text{HurdleLogNormal}(p_P, \mu_P, \sigma) \quad (1)$$

$$C \sim \text{HurdleLogNormal}(p_C, \mu_C, \omega) \quad (2)$$

$$G \sim \text{BetaProportion}(\mu_G, \phi) \quad (3)$$

Differential equations:

$$\frac{d\text{logit}(p_P)}{dt} = e^{b_P} \frac{d\mu_P}{dt} = e^{r_P} \frac{d\text{logit}(p_C)}{dt} = e^{b_C} \frac{d\mu_C}{dt} = e^{r_P} \frac{P}{C} \frac{d\text{logit}(\mu_G)}{dt} = r_G$$

where:

$$r_G = \alpha \quad (4)$$

$$+ \beta_P p_P \log(e^{\mu_P} + 1) \quad (5)$$

$$+ \beta_C p_C \log(e^{\mu_C} + 1) \quad (6)$$

$$- e^\gamma \text{logit}(\mu_G) \quad (7)$$

Initial values:

$$\text{logit}(p_P)(0) \sim \mathcal{N}(-4, 0.5) \quad (8)$$

$$\mu_P(0) \sim \mathcal{N}(-1, 0.5) \quad (9)$$

$$\text{logit}(p_C)(0) \sim \mathcal{N}(-4, 0.5) \quad (10)$$

$$\mu_C(0) \sim \mathcal{N}(-1, 0.5) \quad (11)$$

$$\text{logit}(\mu_G)(0) \sim \mathcal{N}(-1, 0.5) \quad (12)$$

In addition:

- Varying effects across subregions, with spatial GPs for α and $\text{logit}(\mu_G)(0)$. Varying effects have variance τ and GPs have length scale l_{scale}
- Differential equations for P and C estimated piecewise across three time periods: (1) 10,000 BCE to 0 CE, (2) 0 CE to 1600 CE, and (3) 1600 CE to present

Priors:

$$b_P \sim \text{Normal}(-2, 0.5) \quad (13)$$

$$r_P \sim \text{Normal}(-2, 0.5) \quad (14)$$

$$b_C \sim \text{Normal}(-2, 0.5) \quad (15)$$

$$r_C \sim \text{Normal}(-2, 0.5) \quad (16)$$

$$\alpha \sim \text{Normal}(0, 0.5) \quad (17)$$

$$\beta_P \sim \text{Normal}(0, 0.5) \quad (18)$$

$$\beta_C \sim \text{Normal}(0, 0.5) \quad (19)$$

$$\gamma \sim \text{Normal}(-2, 0.5) \quad (20)$$

$$\tau \sim \text{Exponential}(2) \quad (21)$$

$$l_{scale} \sim \text{Exponential}(2) \quad (22)$$

$$\sigma \sim \text{Exponential}(2) \quad (23)$$

$$\omega \sim \text{Exponential}(2) \quad (24)$$

$$\phi \sim \text{Exponential}(2) \quad (25)$$

$$(26)$$

Results